Mapping the Storm

AP Human Geography · Unit 1 · No stimulus (like Question 1) · 7 points · Teacher key · GeoArcade FRQ Tutor

No stimulus. You're interning at a Gulf Coast county's emergency-management office before hurricane season. Your supervisor wants you to master the geographer's toolkit: maps, locations, regions, and how people and the environment shape each other.

A. Identify the type of region that "the South" in the United States is an example of. IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Names it: a perceptual (vernacular) region
- Describes it: exists in people's minds, with fuzzy borders

MODEL ANSWER

"The South" is a perceptual (vernacular) region: it exists in people's minds, so its borders are fuzzy and change depending on who you ask.

Common mistake: "The South is a formal region because it is made up of states." States have official borders, but "the South" doesn't: no law says which states are in it. That makes it perceptual, not formal.

Why the model works: It names the exact type (perceptual, or vernacular) and shows the borders are fuzzy and live in people's minds.

B. Describe how the Mercator projection distorts the size of Greenland compared with the size of Africa. DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Describes the distortion: Greenland looks about as big as Africa
- Gives the truth or the reason: Africa is about 14 times bigger, or Mercator stretches areas near the poles

MODEL ANSWER

T1- The Mercator projection stretches the area of places near the poles, so they look much bigger than they really are.
E2- On a Mercator map, Greenland looks about as big as Africa, but Greenland is about 2.2 million km² while Africa is about 30 million km², roughly 14 times larger.
A3- Because Greenland is near the North Pole and Africa straddles the equator, people who use only Mercator maps can badly underestimate Africa's true size.

Common mistake: "Greenland looks way bigger on the Mercator map." It only says what the map SHOWS. Add the truth (Africa is actually about 14 times bigger) and why (Mercator stretches places near the poles).

Why the model works: It describes the distortion, gives real numbers (2.2 vs. 30 million km²), and explains it with latitude.

C. Describe the difference between a city's site and its situation, using New Orleans or another real city as your example. DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Describes SITE: the physical features of the place itself
- Describes SITUATION: location relative to other places and its connections

MODEL ANSWER

T1- A city's site is the physical land it sits on, while its situation is its location compared to other places.
E2- New Orleans's site is low, swampy land in a bend of the Mississippi River, and about half the city is below sea level. Its situation is near the river's mouth on the Gulf of Mexico, which makes it a port linking the central United States to world trade.
A3- The site explains New Orleans's flood risk, while the situation explains why people built a major city there anyway.

Common mistake: "Site is the city's exact address, and situation is what's going on in the city, like the economy." Two mix-ups: an exact address is absolute location, and situation isn't "what's happening." Site is the physical land; situation is its connections.

Why the model works: It defines both terms, gives proof for each (swampy land vs. a river-to-Gulf port), and shows why the difference matters.

D. Explain how a county government could use remote sensing to respond to a hurricane after it makes landfall. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Says what remote sensing collects: images from satellites, aircraft, or drones
- Explains the response: finding flooding or damage, then sending rescue, supplies, or repairs there

MODEL ANSWER

T1- A county could use remote sensing, which collects images of Earth from satellites, planes, or drones, to see storm damage without sending people into danger.
E2- After a hurricane, before-and-after images show flooded neighborhoods, destroyed homes, and blocked roads; NOAA, for example, flies planes over the coast to photograph hurricane damage.
A3- Knowing where the worst damage is lets the county send rescue crews, water, and generators there first, saving time and lives.

Common mistake: "Remote sensing would help the government know about the hurricane." Too vague. Say what remote sensing collects (images from above) and what the county would do with them.

Why the model works: It says what remote sensing is, what it shows after a storm, and how that changes the response (help goes to the worst areas first).

E. Explain ONE reason a functional region, such as a hospital's service area or a city's commuter zone, can change in size over time. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a cause: new transportation, a competing node, or new technology
- Explains how the edge moves: people come from farther (it grows) or switch to a closer node (it shrinks)

MODEL ANSWER

T1- A functional region can grow when new transportation makes its node faster to reach.
E2- For example, when Washington, D.C.'s Metro Silver Line was extended to Dulles Airport and Loudoun County, Virginia, in 2022, people in those outer suburbs could ride a train into the city.
A3- Because a functional region is defined by connections to its node, faster connections let people come from farther away, so the commuter zone's edge moves outward and the region grows.

Common mistake: "A functional region changes because the government redraws its borders." That's how a formal region, like a county, changes. A functional region has no official border: it changes when connections to its node change.

Why the model works: It names a cause (a new rail line), gives a real example (the Silver Line, 2022), and explains the effect (the region grows).

F. Explain ONE difference between environmental determinism and possibilism, using a real-world example. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Describes determinism: the environment controls what people and societies can do
- Describes possibilism with an example: people choose how to adapt, often with technology

MODEL ANSWER

T1- Environmental determinism says the environment controls how people live and develop, while possibilism says it only sets limits and people choose how to respond.
E2- For example, about a quarter of the Netherlands lies below sea level, but the Dutch built dikes, pumps, and polders (land drained from the sea) to farm and live there.
A3- A determinist would expect such flood-prone land to hold a society back, but the Netherlands became one of the world's wealthiest countries, which supports possibilism.

Common mistake: "Environmental determinism is when the environment determines things, and possibilism is when anything is possible." It just repeats the terms with no example, and possibilism doesn't say ANYTHING is possible: nature still sets limits.

Why the model works: It defines both ideas, gives a real example (the Netherlands), and explains how it supports possibilism.

G. Explain ONE limitation of using only quantitative data, such as census numbers, to understand a place after a hurricane, and how qualitative data could help. EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a limitation: numbers don't show people's reasons, feelings, or experiences
- Explains what qualitative data adds: interviews, field observation, or stories

MODEL ANSWER

T1- One limitation of quantitative data is that numbers show how many people were affected, but not why they made their choices.
E2- For example, the 2010 census counted about 140,000 fewer people in New Orleans than in 2000, after Hurricane Katrina, but it can't show why some families returned and others never did.
A3- Qualitative data, like interviews with residents and field observation, could reveal reasons such as rebuilding costs or ties to family and church, helping leaders plan a better recovery.

Common mistake: “Quantitative data is numbers and qualitative data is words.” Correct definitions, but no answer. Explain what numbers MISS after a hurricane, and how interviews or field observation fill that gap.

Why the model works: It names what numbers miss (the why), gives a real example (New Orleans after Katrina), and explains what interviews would add.

Scoring guide written for the GeoArcade FRQ Tutor. Award each part independently, 1 point each.